

Module A1 Summative Assessment

Evidence backed ecosystem mapping through AI assisted statistical work

Module OS-A01 | Asset OS-A01-A09 | Canonical lesson OS-A01-A01

Assessment purpose

This summative assessment determines whether you can produce a defensible national information ecosystem map and professional recommendation. You must demonstrate the underlying statistical task, decide whether and how AI should assist, verify every consequential output, correct failures, compare quality-adjusted effort and retain accountable human judgement.

Assessment conditions

Control	Requirement
Assessment mode	Individual evidence dossier followed by an oral defence.
Working time	Up to 180 minutes for the dossier and 15 minutes for the oral defence.
Permitted resources	Canonical lesson, A07 synthetic data, approved notebook, approved AI environment and cited authoritative references.
AI use	Permitted and expected for bounded assistance. Every prompt, configuration and selected output must be retained.
Human authority	You remain responsible for statistical validity. Domain claims requiring specialist authority must be identified for concurrence or escalation.
Submission	One evidence dossier containing all mandatory components listed below.

Assessment scenario

The Republic of Kintaba requires district crop information six weeks before the annual probability survey estimate. Leaders are considering administrative reporting, registry data, weather observations, geospatial evidence and AI-assisted synthesis. The evidence register contains authority, coverage, quality, user-need and evaluation statements. Your task is to map the relevant ecosystem and recommend whether one bounded AI-assisted workflow should proceed, be restricted, be redesigned or stop.

Required professional product

- A bounded ecosystem map covering the decision, population, product, actors, information assets, flows, users and authority.
- A baseline workflow and recorded elapsed effort for a comparable conventional task.
- A documented AI task-allocation decision and versioned prompt or configuration.
- The preserved AI candidate output, claim ledger, independent verification and correction log.
- A fitness-for-use and total-error analysis, including coverage and generalization limits.
- A quality-adjusted time comparison that includes prompting, checking, correction and rework.
- A professional recommendation with conditions, stopping rules, residual uncertainty and named human accountability.

Critical evidence rule

A polished final map is insufficient without process evidence. Unsupported AI claims, unapproved information exposure, AI self-verification, fabricated evidence or missing human accountability are critical failures regardless of the numerical score.

Task 1 Statistical purpose and ecosystem boundary

Define the decision need before using AI. State the intended use, target population, statistical product, geography, reference period, quality priorities, consequence of error and explicit exclusions.

Field	Candidate response
Decision and user	
Target population	
Statistical product	
Geography and period	
Quality priorities	
Consequence of error	
Boundary and exclusions	

Task 2 Conventional baseline

Without AI assistance, extract and structure four evidence-backed actors or assets. Record the start and end time, ambiguities encountered and acceptance checks used.

ID	Actor or asset	Role or meaning	Evidence ID	Accepted

Baseline field	Record
Elapsed working time	
Ambiguities and rework	
Acceptance criteria applied	

Task 3 AI allocation and task specification

Required field	Candidate response
What AI could do	
What AI should do	
What AI must not do	
Approved inputs	
Required output schema	
Evidence requirements	
Statistician duties	
Subject specialist duties	
Stopping rules	

Versioned prompt or configuration

Field	Record
Version	
Tool or capability class	
Prompt or configuration	
Reason for revision	

AI interaction timing

Activity	Elapsed time
Task specification and prompting	
Generation or tool execution	
Total before verification	

Task 4 Candidate output verification and correction

Claim ID	AI candidate claim	Evidence or test	Finding	Disposition

Correction log

Record	Material error	Cause	Correction	Human reviewer

Verification effort

Activity	Elapsed time
Source and authority checks	
Statistical and reproducibility checks	
Specialist consultation or escalation	
Corrections and rework	

Task 5 Fitness for use and ecosystem assurance

Dimension	Evidence based judgement	Residual limitation	Human authority
Relevance			
Accuracy			
Reliability			
Timeliness			
Coherence			
Comparability			
Accessibility			
Interpretability			

Total error and integration risks

Risk	Evidence	Effect on use	Control or escalation
Coverage			
Measurement			
Nonresponse or missingness			
Processing			
Linkage or duplication			
Generalization			

Task 6 Quality adjusted efficiency

Effort component	Conventional method	AI assisted method	Evidence or comment
Task framing			
Extraction or structuring			
Verification			
Correction and rework			
Specialist review			
Total elapsed effort			

Decision field	Candidate conclusion
Did both outputs meet the same quality threshold?	
Net time or effort saved	
Risk or rework that offset the gain	
Reusable prompt pattern or procedure	
Quality-adjusted efficiency conclusion	

Task 7 Professional recommendation

Decision element	Candidate recommendation
Disposition proceed restrict redesign or stop	
Permitted AI role	
Nondelegable statistical decisions	
Required specialist concurrence	
Verification before operational use	
Information protection conditions	
Stopping rules	
Residual uncertainty	
Named accountable authority	

Submission checklist

Required evidence	Included	File or section reference
Ecosystem boundary and map		
Conventional baseline and timing		
AI allocation decision		
Prompt or configuration versions		
Preserved candidate output		
Claim ledger and independent checks		
Correction log		
Fitness-for-use analysis		
Quality-adjusted efficiency comparison		
Professional recommendation		
Reusable controlled artifact		

Candidate declaration

I confirm that the submitted evidence distinguishes my professional judgement from AI-generated content, records the tools and sources used, contains no knowingly fabricated evidence, and identifies decisions requiring statistical or subject-matter authority.

Field	Record
Candidate name and institution	
Assessment date and assessor	
Approved AI environment	
Dataset version	OS-A01-A07 v1.0
Candidate signature	
Declared limitations	

Oral defence preparation

- Identify one AI output you rejected and cite the evidence.
- Explain one correction that materially changed the map or recommendation.
- Defend the quality-adjusted efficiency conclusion.
- State which decision cannot be delegated to AI and where specialist concurrence remains necessary.